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Foot shape analysis of professional American Football players

Gwansik Park[a] and Richard Kent[a,b]

[a]Biomechanics Research and Consulting LLC, Charlottesville, VA, USA; [b]Center for Applied Biomechanics, University of Virginia, Charlottesville, VA, USA

ABSTRACT

The goal of this paper is to quantify the shape of National Football League (NFL) players' feet and to identify foot shape characteristics that facilitate the development and refinement of position-specific footwear for NFL players. Statistical shape analysis was performed on three-dimensional (3 D) scans of 290 NFL players' feet obtained from 4 NFL teams during preseason training camps to identify the principal components that explain characteristics of the shape variability. Eight foot shape measurements were then defined to reflect the main principal components and were quantified. The foot volume, the arch shape in conjunction with the foot length, and the foot curvature explained more than 80% of the foot shape variation. The average foot length was 293 ± 13.7 mm and width was 110.6 ± 6.8 mm. Significant differences were observed between position groupings, primarily with regard to size. The prevalence of flat feet (defined as 0 arch depth) was greater in linemen and hybrid position groups than in the skill group. This information can facilitate improvements in the lasts used to create shoes for elite-level players and can also help inform footwear selection.

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Statistical shape analysis; foot shape; 3D foot scanning; NFL players; American football

Introduction

The National Football League (NFL) has recently implemented a footwear sizing system that utilises three-dimensional (3 D) scanning to quantify a player's foot shape and an algorithm to match it geometrically to the interior dimensions of available footwear (Wannop et al., 2019). During the development of the system, a player foot scan database was created by scanning 1451 NFL players. This paper analyzes a sample of those scans to describe the shapes of NFL players' feet. Understanding foot shape and its variation can inform the design and selection of footwear, particularly for elite athletes, since fit can affect comfort (Goonetilleke & Luximon, 2001), performance (Caselli, 2006), and the risk of foot injuries such as fifth metatarsal fractures (Mayer et al., 2014), foot deformities (Luximon, 2001), turf toe and Lisfranc injuries (Frimenko et al., 2012), and blisters (Knapik et al., 1999). A comprehensive review of footwear fitting, sizing, and scanning literature was recently published by Wannop et al. (2019).

Several methods have been implemented to describe foot shape variation in humans. For example, Kouchi (1995) analysed the variation in foot outlines using the flexion angle of the medial axis of the foot outline. Luximon (2001) analysed foot shape data of 50 individuals using 25 anthropometric measures. The author concluded that foot length, width, height, and curvature were key shape parameters, and explained 77% of the foot shape variation. Rodrigo et al. (2012) used a turning function to characterise the outline of the foot in the transverse plane at the heights of 2 and 40 mm from the bottom surface, and classified four different types of feet. More advanced 3 D foot shape characterisation has been done using statistical shape analysis (SSA), which encompasses a range of approaches to analyse shape variability based on dimensionality reduction and orthogonality (for example, principal component analysis). Baek and Lee (2016) proposed an SSA framework for foot shape based on a database of 3 D scanned feet. They utilised a clustering technique with foot landmarks extracted from foot shapes for the SSA.

CONTACT Richard Kent rkent@biocorellc.com
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The current paper expands on these earlier studies both methodologically (SSA of the entire 3 D surface instead of either a 2 D outline or landmarks) and in the specificity of the population studied. The NFL player population is of particular interest for such an analysis. Human activity can generate substantial alterations in foot shape (Wannop et al., 2019), and a population of elite athletes who are large and perform unique activities would be expected to exhibit unique foot shape attributes. Furthermore, elite athletes have access to sport-specific, level-specific, position- (or body size-) specific, and even fully personalised footwear.

Methods

A stepwise process was employed to generate a distribution of position-specific foot measurements from the scan data. First, a sample of foot scans was extracted for processing. These scans were cleaned and aligned. A statistical shape analysis (SSA) was then performed to identify the number of principal components that explained the key shape attributes in players' feet. A group of eight measurements was then identified to be (1) Sufficient for describing the key principal components identified by SSA, (2) Measurable in a foot scan, and (3) Definable relative to anatomical landmarks or current foot measurements used for footwear development and selection (e.g. foot length, foot width, and arch length). The scans were then binned by nominal position groups (linemen, hybrid, skill) and the distribution of foot measurements was determined for each group. The details of each of these steps are described below.

Foot scan database of the NFL players

A footwear fitting system has been deployed in the NFL to identify footwear that best fits a player's foot (Wannop et al., 2019). During the development of the system, three-dimensional (3 D) foot scans of 1451 NFL players were obtained from all 32 NFL teams during preseason training camps. A subset of 290 left-foot scans from four teams was extracted for analysis. The average weight (111.8 ± 21.5 kg) and height (187.8 ± 6.9 cm) of the subsampled population were comparable to the population of NFL players ($N = 11{,}795$) during the 2015–2018 seasons (weight: 111.0 ± 21.0 kg, height: 188.1 ± 6.7 cm) from NFL Game Statistics and Information System (GSIS) database (Dreyer et al., 2019; Lessley et al., 2018).

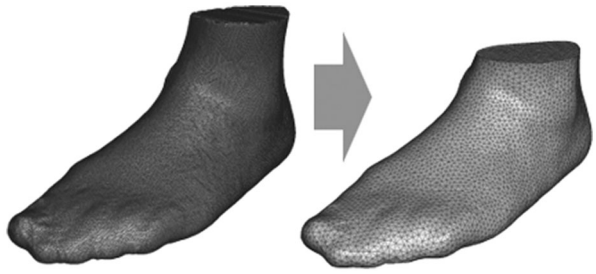

Figure 1. Example of foot scan data prepared for the SSA: raw scan data (left) and prepared data (right).

Table 1. Category of player's position.

Group	Position
Linemen	DE, DT, LB, NT, T, C, OG, OL, OT
Hybrid	LB, MLB, OLB, TE
Skill players	CB, DB, FS, SS, S, QB, RB, WR

Scans were collected by experienced users with a commercially available laser foot scanner (INFOOT USB model IFU-S-01, I-Ware Laboratory Co., Ltd., Osaka, Japan) connected to a Windows-based PC that collected data at 1-mm resolution in an ASCII geometric database file format (FBD). The players were instructed to step carefully onto the plate of the scanner with one foot, the other foot on the support step. Foot scans were imported into Meshmixer v3.5 (Autodesk, San Rafael, CA), and prepared for SSA; the size of the faces of the scan data was reduced to a nominal value of 3.5 mm, irregular surfaces were cleaned-up as necessary using a smoothing function, and the scans were sectioned at the transverse plane immediately proximal to the superior aspect of the medial malleolus. The geometry distal to that plane was retained as the foot shape for the SSA (Figure 1). The error (defined as the minimum distance between the original surface to the cleaned-up surface) from this cleaned-up procedure was 0.3 ± 0.6 mm across 290 samples.

The scans were categorised into three groups by player position: linemen (83 scans), hybrid (68 scans), and skill (135 scans) players (Table 1). Kickers and punters were not included in this study due to the limited number of scans available.

Statistical shape analysis (SSA)

Statistical shape analysis is a technique to identify shape attributes that best represent a larger number of data sets based on a dimensionality reduction analysis, such as principal component analysis (PCA). PCA returns principal components (PCs), which indicate a new basis to best represent complex data, and their variances, which can be interpreted as the measure of the importance of each principal component. The PCs are ranked in descending order according to their variances, therefore, for example, the first PC is the most important basis (i.e. it describes the most shape variability).

The first step to conducting the SSA was to ensure a correspondence between foot shapes in such a way that the location at any given point in one shape model could be mapped to a corresponding point on another model (Bryan et al., 2010). First, the foot scans were aligned with the foot coordinate system (FCS), which was defined with the x-axis along a line between the tip of the heel and the tip of the second toe, and the xy-plane as the bottom surface of the foot (Figure 2). 240 pseudo-landmarks (Figure 3) were then obtained from eight transverse cross-sectional contours of the foot (three anatomical sections and five pseudo sections). Details on the method to select the sections and pseudo-landmarks are in the Supplementary Appendix.

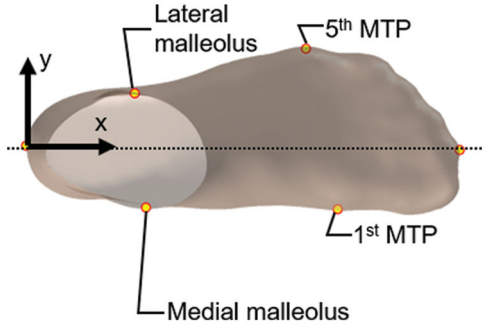

Figure 2. Foot coordinate system (FCS) and anatomical landmarks.

Correlation-based PCA was performed using the function 'PCA' in the statistical toolbox in MATLAB (R2017a, The MathWorks Inc., Natick, MA). The correlation matrix for PCA was constructed with 240 pseudo-landmarks selected from the foot scans ($N = 290$). The percentage variance of each PC describes its contribution to the total shape variation.

To visualise the principal components and to generate the average foot shape of the player population, 3 D foot shapes were reconstructed implementing a thin-plate spline with a radial basis function (Rohr et al., 2001). This morphing technique transforms a template foot shape (a randomly selected sample among 290 samples) to the target shape using the 240 pseudo-landmarks selected for the SSA. The error (defined as the minimum distance between the morphed surface to the target surface) from this morphing procedure was 0.63 ± 0.17 mm across 290 samples (Figure 3).

Foot shape measurements

Eight foot measurements were defined to quantify the foot shape variation across the NFL player population (Table 2, Supplementary Figure A2). These measurements, which were selected to describe the PCs identified by the SSA, were binned by position groups and reported as distributions within those groups.

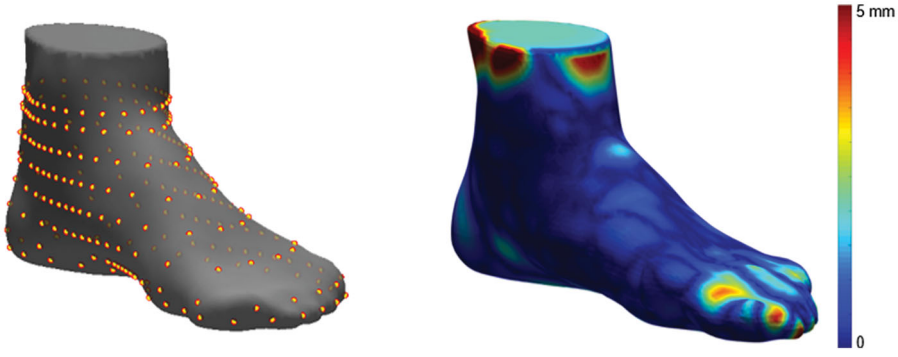

Figure 3. 240 Pseudo-landmarks for the SSA (left) and reconstruction error using the morphing technique as an example from one of the samples (right).

Table 2. Interpretation and definition of foot shape measurements (Supplementary Figure A2).

Measurement	Interpretation and definition
Foot length	A distance from the tip of the heel (pternion) to the tip of the longest toe, taken along the *x*-axis of the FCS
Foot width	The maximum horizontal breadth across the foot along the *y*-axis of the FCS
Arch length	A distance from the heel tip to the first MTP along the *x*-axis of the FCS
Heel width	The maximum horizon breadth (along the *y*-axis of the FCS), measured from the cross-sectional contour on the heel width plane*
Mid-foot width	The maximum horizon breadth (along the *y*-axis of the FCS), measured from the cross-sectional contour on the mid-foot plane*
Ball girth	A circumference of the foot, measured from the cross-sectional contour on the ball girth plane*
Arch depth	The maximum perpendicular distance between the line across the first MTP and the medial heel and the cross-sectional contour of the foot on the footprint plane*
Flare angle	Measurement of the curvature of the foot, defined as the angle along the *z*-axis in FCS between the midline of the heel (defined as the line across the heel to mid-point of the heel width) and the midline of the whole foot (defined as the line across the heel to the mid-point of mid-foot width)

*The details on the definition of cross-sections used for defining the foot shape measurements are provided in Appendix (Supplementary Table A1).

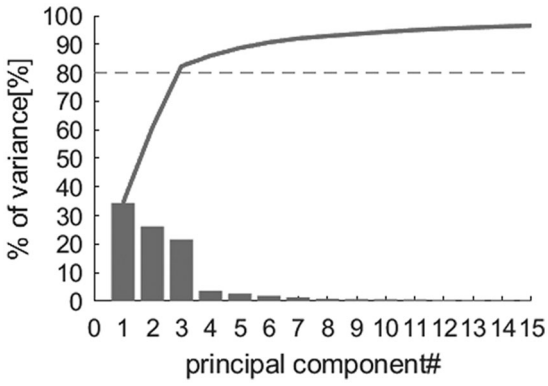

Figure 4. Percentage of variance explained by principal components: bar graph explains % of the variance, and the solid red line is a scree plot of the % of the variance.

Results

The results of the SSA showed that the first three PCs explained 82.3% of the shape variation in the players' feet (Figure 4). The first PC is related to the volume (size) of the foot (34.4%), and hence the length, width, and girth (Figure 5). The second PC is related to the medial longitudinal arch shape of the foot in conjunction with the foot length (26.2%) and the third PC reflects the curvature of the foot (21.7%) (Figure 5). Videos describing the first three PCs and the representative foot shape of the NFL players in digital stereolithography (STL) format is included as supplementary material to this paper.

Table 3 summarises the foot measurements that were extracted as descriptors of the first three PCs and binned by position group. All measurements except the arch depth and the flare angle were significantly different ($p < 0.05$) between the position groups. Histograms of all foot measurements and raw data including foot measurements, player characteristics, and positions are provided in the Supplementary Appendix.

Although the difference in the arch depth between position groups was not statistically significant, the players in the linemen and hybrid groups showed a larger percentage of players with zero arch depth (i.e. flat feet) than the skill players (Figure 6).

Discussion

This paper analysed the foot shapes of NFL players using SSA and quantified eight foot measurements intended to describe the PCs, which reflects the key shape attributes in players' feet. Key measurements identified in this study included measurements typically related to foot size (e.g. length and width), as well as the arch shape and the flare angle, which help to capture the shape attributes due to the second and third PCs (the arch shape and the foot curvature). These findings are consistent with Luximon (Luximon, 2001), who claimed foot length, width, height, and curvature explain 77% of the foot shape variation, and others who have identified the arch shape as one of the main parameters to classify foot shapes (Dahle et al., 1991; Rodrigo et al., 2012; Xiong et al., 2010). Our findings are also consistent with Xiong et al. (Xiong et al., 2010), who found that longer feet tend to associate with lower arch height. These findings point to limitations of the Brannock device (Wannop et al., 2019), which has conventionally been used for

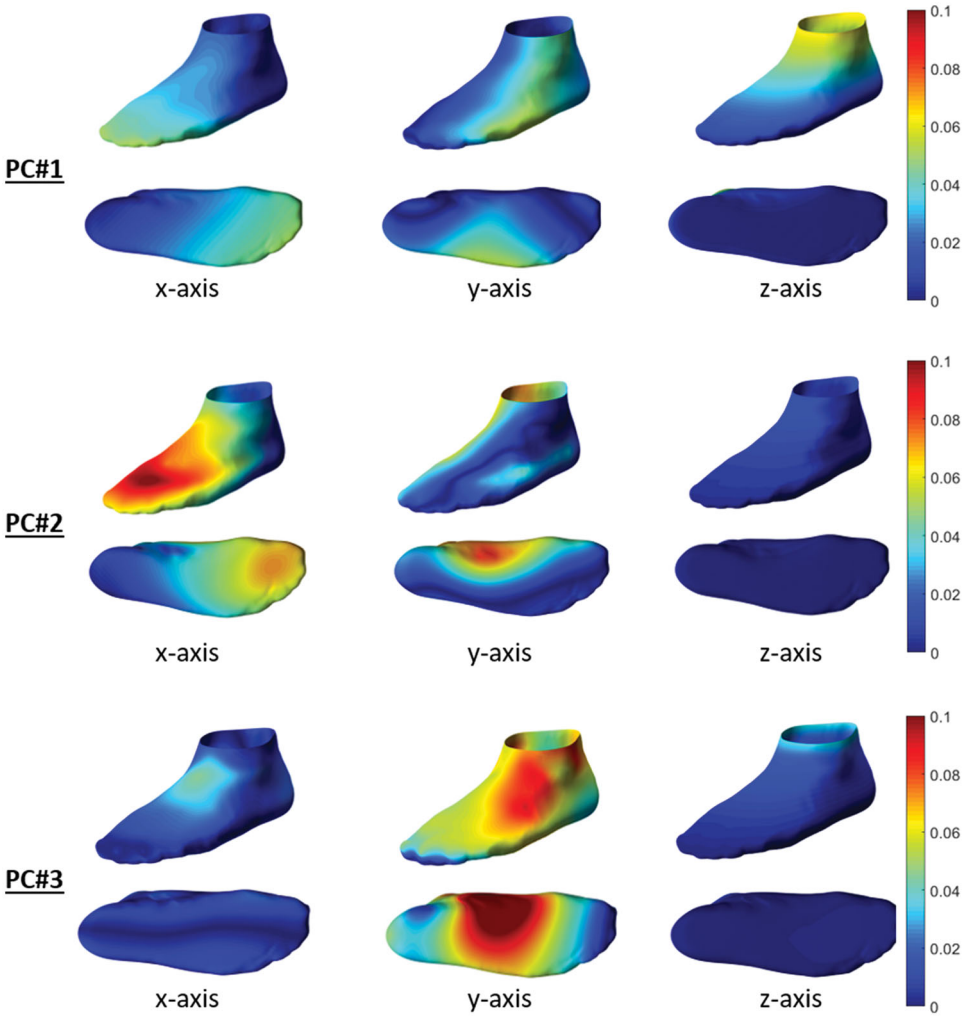

Figure 5. Visual representation of the first three PCs: deformation contour between the mean and the +2.0 standard-deviation (σ) shape model normalised by the maximum foot dimension in each direction of the FCS.

Table 3. Foot shape measurements of NFL players.

Measurements	All players ($n = 290$)		Position group: Linemen ($n = 83$)		Hybrid ($n = 68$)		Skill players ($n = 135$)	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Foot length [mm]	293.4	13.7	303.7	11.3	297.7	10.4	285.5	11.0
Foot width [mm]	110.6	6.8	117.0	5.9	111.3	4.5	106.5	5.1
Arch length [mm]	210.7	9.8	218.1	8.1	213.8	7.5	205.0	7.9
Heel width [mm]	70.1	4.8	74.8	4.3	70.5	3.4	67.5	3.9
Mid-foot width [mm]	100.8	7.3	106.8	7.3	101.9	5.5	96.6	5.2
Ball girth [mm]	262.8	15.0	277.0	12.4	265.3	9.9	253.2	10.5
Arch depth [mm]	14.7	8.7	14.5	9.1	15.4	10.5	14.5	7.3
Flare angle [deg]	1.4	2.2	1.1	2.1	1.6	2.6	1.4	2.0

selection of footwear, but does not measure arch shape or foot curvature directly.

The key contribution of this study is the consideration of a specific population of elite athletes that have special footwear requirements. Well-designed footwear can improve athletic performance and reduce the risk of certain types of athletic injury, but these benefits can be mitigated by improper fit. Thus, proper fitting footwear, especially at an elite level, is paramount. The dimensions reported in this paper should be considered in the construction of foot lasts (Rodrigo et al., 2012) and footwear intended for use in the NFL. Specific information such as that reported in this paper is preferred over template scaling or other less-specific methods of developing foot lasts (Wunderlich & Cavanagh, 2001). Furthermore, the methodology developed here may provide a framework for quantifying foot shape among other populations that may have specific footwear sizing requirements.

At elite levels of football, shoes are generally designed for specific categories of player (e.g. by player position and weight). The finding of increased flat foot in the linemen and hybrid groups from this study supports development

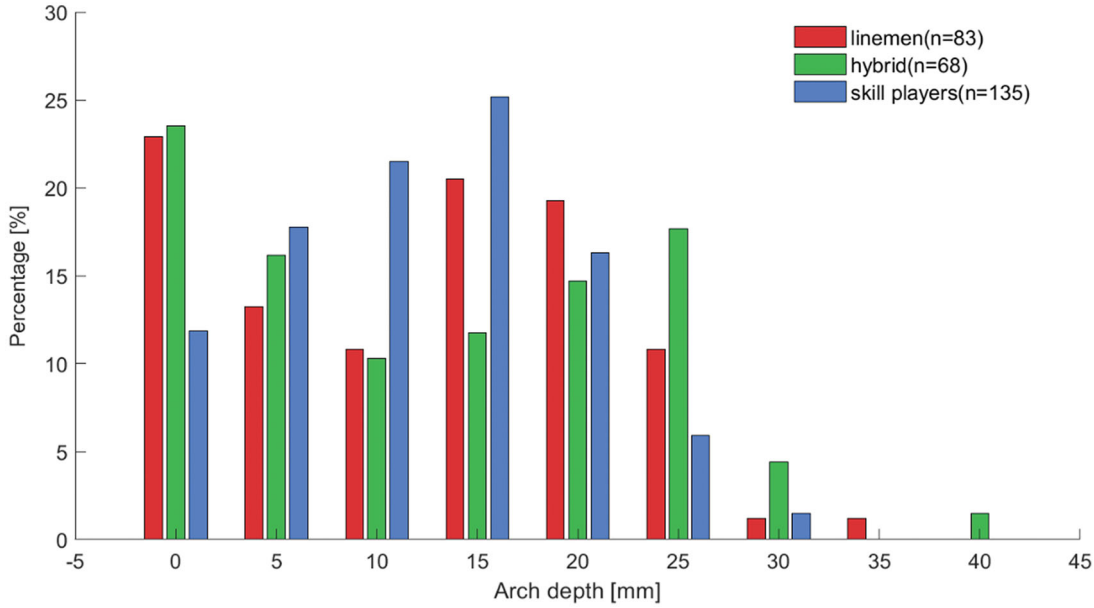

Figure 6. Histogram of the arch depth for three position groups: linemen (red), hybrid (green), and skill players (blue).

towards even more specificity, particularly given the link between arch geometry, arch mechanics, and possibly injury incidence (Williams & McClay, 2000) and the increased focus on footwear as a piece of protective equipment, not simply an extension of the uniform apparel (Jastifer et al., 2017).

Conclusion

Understanding the foot shape variation among NFL players is important to improve the design and selection of footwear for comfort, performance, and injury mitigation. This study identifies the critical principal components (foot volume, arch shape, and foot curvature) and quantifies eight foot measurements (foot length, width, arch length, heel width, mid-foot width, ball girth, arch depth, and flare angle) for position groups within the NFL player population. This information can facilitate improvements in the lasts used to create shoes for NFL players and can also help inform footwear selection.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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